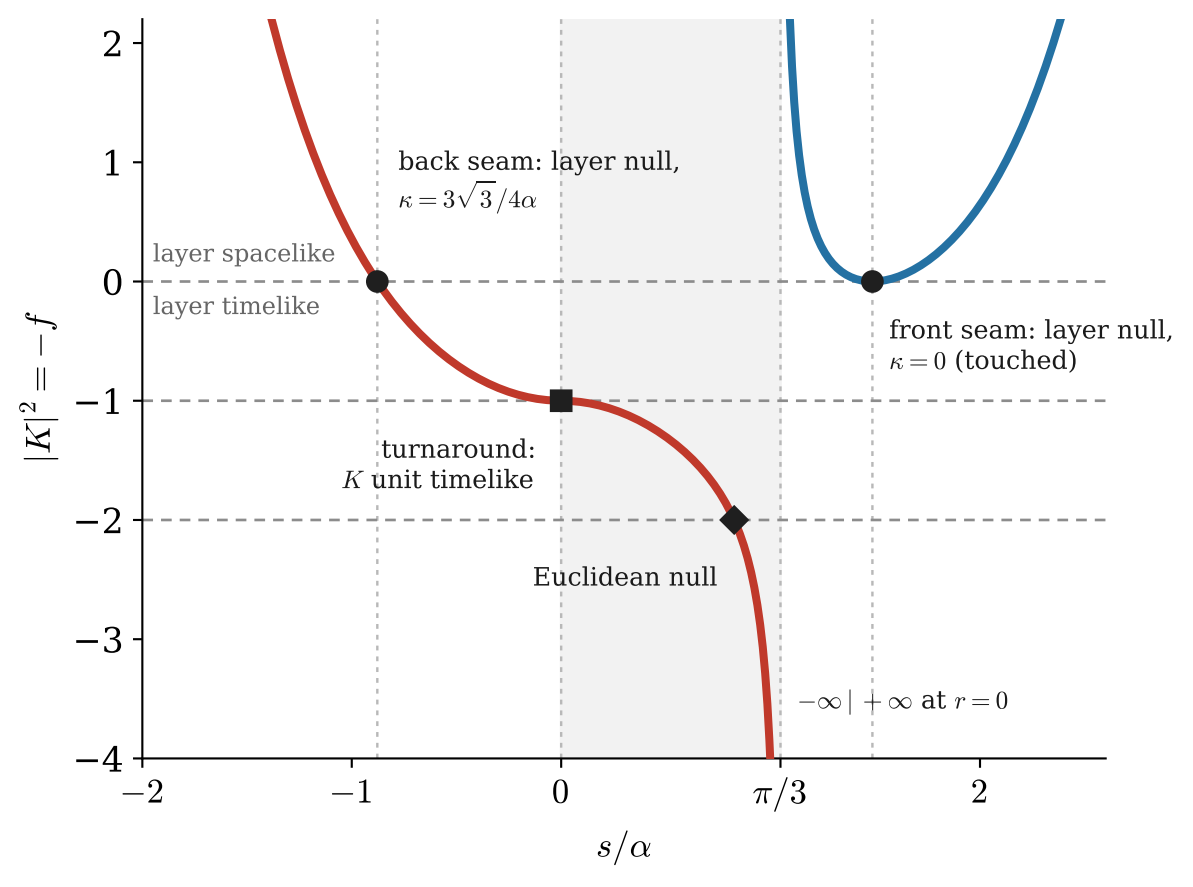
back seam: layer null,
$\kappa = 3\sqrt{3}/4\alpha$
layer spacelike
layer timelike
front seam: layer null,
$\kappa = 0$ (touched)
turnaround:
K unit timelike
Euclidean null
$-\infty \,|\, +\infty$ at $r = 0$
$|K|^2 = -f$
s/α
2
1
0
−1
−2
−3
−4
−2
−1
0
$\pi/3$
2